



System Acceptance Review
University Rover Challenge 2025



BRAC University
BRACU Mongol-Tori

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1. Introduction:

The team BRACU Mongol-Tori from BRAC University consists of 50 undergraduate students from various engineering disciplines. The team comprises seven technical sub-teams dedicated to contributing to core engineering segments of the rover and URC missions. Team's Research and Documentation sub-team engages undergraduate students in publishing research papers based on the innovation done by the rover. On the other hand, the Outreach and Finance sub-team influence high school students in robotics and STEM education. The team utilizes a concurrent engineering model to prepare the rover "Hypersonic" for the 2025 competition.

2. Rover Core system:

Owing to past experiences and failures, the team opted for a better system design that is shown in the **Figure - 4** to mitigate any technical errors. A partial redesign of the mechanical system along with the control system and the autonomous system has been made, keeping innovation, redundancy, and modular design.

2.1 Mechanical:

2.1.1 Chassis:

The rocker-bogie suspension system has proven highly effective for navigating rough terrain, and as such, *Hypersonic* is equipped with an advanced arched base support integrated with this system. This system allows the rover to distribute the weight efficiently and maintain stability even on steep slopes or rocky fields. The chassis is segmented and can be assembled and disassembled according to mission requirements. A flipping mechanism is integrated into the chassis, making a separation in between the controls and the electronics systems to enhance accessibility during maintenance, troubleshooting, and debugging. A Jamdani-inspired CNC-machined design on the rover's aluminum panels enhances aesthetics, reduces weight, and preserves structural integrity while honoring Bengali culture.

2.1.2 Manipulator:

The *Hypersonic*'s 6-DOF manipulator is designed to assist the driver in completing all tasks with precision. The arm operates within a truncated spherical sector with a 1.4-meter radius and a 120° rotational base, allowing access to any point within this 3D workspace. For the Science Mission (SM), *Hypersonic* swaps out its standard gripper with an excavator claw. Additionally, a 3D-printed adaptive gripper will be added to provide a resistive and secure grip. A clip-on system enables the attachment of an autonomous keyboard-typing arm, which is installed only during Equipment Servicing (ES). The 2-DOF keyboard-typing manipulator autonomously extends to a predefined position upon deployment to initiate typing. A fixed camera machine vision algorithms are used to calibrate with selected reference markers on the keyboard for self-localization, enabling precise navigation to each key using inverse kinematics. As the key coordinates are fixed, the system can accurately determine and execute the required movements, allowing *Hypersonic* typing autonomously.

2.2 Electronics:

The rover is powered by three hot-swappable 18V 9Ah lithium-ion batteries, each independently supplying power to the control and communication systems as well as peripherals. A dual channel Cytron 60A Bi-Directional DC Motor Driver, is used for the wheels of the rover, and four dual channel 10A Cytron MDDS10 are used for the arm, keyboard-typing manipulator's extender, and one channel is kept for backup in case any channel needs swapping. The Cytron motor drivers are stacked on top of one, represented in **Figure - 3**, for making the control system compact. Furthermore, in case of any

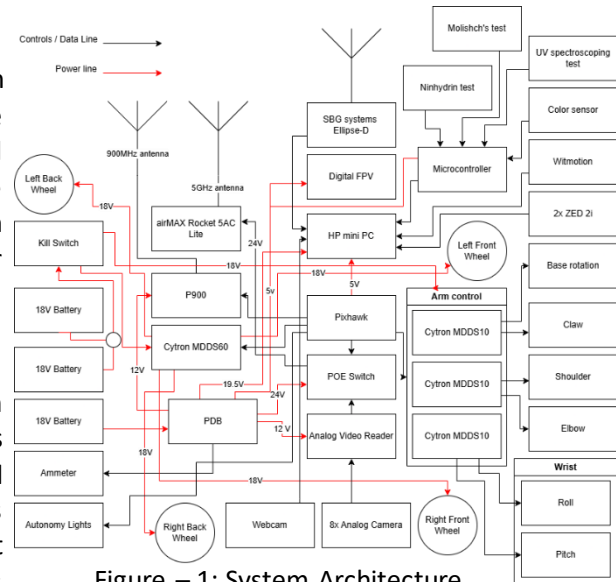


Figure – 1: System Architecture

intervention, it has been made easier to debug each channel separately using the onboard power monitor. The estimated runtime of the rover is 90 minutes. During this runtime status LEDs are controlled with Pixhawk as per mission tasks powered by 12V relays. The vision module operates at 12V and 5V for analog cameras and digital FPV respectively. The communication module includes long-range telemetry module (P900) which is powered by a 12V supply, while the 5GHz communication router receives regulated 24V from an edge router PoE switch. The onboard computing unit, an HP Mini PC, requires a 19.5V power input for operation. The Power Distribution Board (PDB) ensures stable power delivery to all components, optimizing system reliability and efficiency. Furthermore, a red-labeled physical kill switch controlled by a 12V relay with reverse polarity protection using electronic switch is implemented for critical safety measures.

2.3 Software:

Hypersonic utilizes the Robot Operating System 2 (ROS2) Humble framework for both teleoperation and autonomous navigation (AN). The rover is controlled using mission-specific ROS packages that communicate via the MAVLink protocol, ensuring lossless and uninterrupted communication. To improve location accuracy despite weather constraints (e.g., cloud cover), alternating Global Navigation Satellite System (GNSS) modules are used to determine precise latitude and longitude. To eliminate the possibility of magnetometer drift during mission, combinations of industry-grade Inertial Navigation System (INS) are used, which are swapped when the drift is detected to the autonomous topic. Home brewed Graphical User Interface (GUI) also enables users to visually detect, adapt, and view the rover's status at all times. For the AN, the user will be able to input any GNSS and select the detection method for the rover to complete that task. The science GUI enables the user to visually inspect the analysis performed by the science module depicted in **Figure - 5**. It also includes tools and a detection process for potential life samples, as required for the competition. A combination of advanced data visualization and processing techniques is used to support critical analysis.

2.4 Network and Vision:

The rover utilizes a redundant communication system for point-to-point communication with the capability of frequency hopping and load balancing. The rover communicates with the base station using 900MHz Holybro P900 telemetry and a 5GHz Ubiquiti airMAX AC Lite router along with a 13dBi airMAX omnidirectional (AMO-5G13) antenna. The rover also uses a dedicated Walksnail Avatar HD kit V2 for long-range video transmission while in line of sight. The outdoor antenna setup comprises a 5GHz 19dBi 120° sector antenna (AM5G19-120), a 900MHz omnidirectional antenna, and a 5GHz 180° Walksnail Avatar FPV VRX antenna for FPV. This whole system is powered by a single 18V lithium-ion battery, which has the capability of powering the whole setup for 4 hours. This setup is tested by placing it at 3 meters in height and 25 meters away from the command station. Furthermore, 8 analog cameras and a Logitech C922 webcam are used for vision. The analog feeds, along with the webcam, are processed onboard, encoded to Real-Time Streaming Protocol (RTSP), and transmitted via a 5GHz network. In case of interference, the communication modules hop to other unlicensed frequencies maintaining rover to base uninterrupted communication.

2.5 Autonomous:

For the Autonomous Mission (AM), the rover utilizes a tactical grade INS, SBG Ellipse-D, integrated with dual antenna Differential GNSS (DGNSS) and Real-Time Kinematics (RTK) positioning, ensuring centimeter accuracy in single-point and 0.2-degree heading precision. A Witmotion HWT905, industry-grade Altitude Heading Reference System (AHRS) Inertial Measurement Unit (IMU), is used as a secondary IMU, which ensures 1-degree heading accuracy in a static magnetic field. This dual IMU system, enhanced by DGNSS, is used to calculate the target heading and align itself toward the destination. Rover utilizes two ZED 2i stereo vision cameras for depth mapping and distance calculation for detecting the object of interest. For object recognition, the rover integrates the YOLO11 and YOLOv8-world models trained on our custom dataset of mallets and bottles, respectively.

3. Approach to Mission Plan:

3.1 Delivery Mission:

To complete the Delivery Mission (DM) the rover has been designed and developed to find and pick up objects, read signs, and deliver the sample to astronauts by traversing distant paths in adverse terrain. From the previous year's competition experience, the linear actuator-based omni-wheel mechanism has been removed because of its time of deployment. This also resulted in an increase in the rover's stability and capability of traversing precisely through marked paths. The 12-inch-diameter slotted wheels give the ability to traverse through sandy terrain. The TPU-based 3D-printed adaptive end effector shown in **Figure - 2** allows the rover to pick up any tools, rocks, and lid of any heavy object of more than 5 kg in mass. It can also open the hinged lids of the toolbox. The adaptive feature of the claw also enables it to hold any object that is greater than 7.5 cm in length. The base rotation of the arm allows it to store any sample objects in the cache box of the rover. The onboard cameras are encoded in RTSP and streamed through 5GHz communication modules to get rover feed and tested over 1km on a direct line of sight. In case of any network failure, the rover utilizes a dedicated digital FPV whose transmitter will be lifted up calculating the Fresnel Zone just right before the start of the mission. The SBG INS ensures uninterrupted operation, and a redundant control system enables Pixhawk to take over in the event of a power failure, allowing the rover to seamlessly complete Stage 1 and Stage 2 without the need for battery replacement.

3.2 Equipment Servicing Mission:

This year, the manipulator of the hypersonic has been designed to perform various maintenance tasks on the mock-up lander in a precise way. Using the same manipulator of DM, a dedicated servo-based mini-arm lets the rover perform autonomous keyboard typing operations. A motorized rail is used to position the arm towards the keyboard while a solenoid is used to press the key. Among the 8 cameras, 3 of them are placed in the arm along with a laser crosshair for precise operations. This allows the rover to pick and place cache, remove and insert USB (type A), push buttons, and flip switches. The base rotation has been implemented to operate the joystick in all 4 positions. To perform the task of turning knobs, bevel gear has been implemented. The rover has gone through multiple mock tests and is capable of doing these tasks within 30 minutes over 0.25 km.

3.3 Autonomous Mission:

The rover's autonomous navigation system is built on a custom decision tree which takes GNSS goals as inputs. The decision tree handles switching between path planning, navigating, searching, and arrival. This year we have built a custom path planning algorithm that uses the point cloud map of the environment to determine the elevation of terrain and planning a shortest path. The global planner of the decision tree uses a point-to-point follow algorithm while continuously updating heading alignment of the rover and a local planner maps the terrain, for dynamic obstacle avoidance, until it reaches the required distance for stopping the rover. The processing of the navigation stack is done on an onboard mini computer, while the communication between different sensors is done by ROS2 nodes. After navigating to the given GNSS coordinates of an object or ArUco post, the navigation system switches to its searching mechanism. For the searching of ArUco posts, the frames captured from two ZED 2i stereo cameras are stitched together for an ultra-wide Field of View (FOV) of 200 degrees. The detection part of the ArUco markers is done by the OpenCV ArUco library, and ZED cameras are used to calculate the marker distance from 15 meters away. The detection period is set to 2 seconds, if the rover's camera immediately does not find any ArUco post within this time period, the rover initiates a square spiral search pattern, where it rotates in 360 degrees with 60-degree increment and is followed by a minimal time delay. This procedure mitigates the effects of the vibrations caused during rover rotation. After completing a 360-degree rotation, it goes to a vertex of the square pattern and repeats the same procedure until it finds the desired ArUco post. During autonomous traversal, the rover will turn on a visible red light and will flash a green light when it reaches a GNSS coordinate or reaches any post or object. The object detection models were trained on a custom dataset consisting of 6330 labeled images of orange mallets and water

bottles at varying distances, angles, and terrains. The model was hyperparameter tuned for faster inference while achieving higher accuracy. Distances and directions of the objects are determined by perspective, transforming the bounding boxes from the inference. All of this can be monitored from our base station with a web Graphical User Interface (GUI), which shows the real-time position of the rover and camera feed in an offline map. GNSS coordinates can be entered from the web GUI or ROS2 interface. In the event of an issue, intervention can be executed from the GUI and use teleoperation to take control of the rover.

4. Testing and Evaluation:

Mongol-Tori follows a comprehensive testing procedure to ensure optimal Rover functionality.

a. Sub-System Testing: The sub-system testing and development phase runs from September 2024 through February 2025, with different components tested at various stages. The Computer Aided Design (CAD) designs and simulations for mechanical structures, science modules, arm, and wheels are being developed between September 2024 and February 2025. Network systems tests are started from January 2025, including implementations of 5 GHz, FPV, and 900 MHz communication. Camera testing is scheduled from November 2024 to January 2025. Electrical systems and power units are being tested from November to December 2024.

b. Complete System Testing: Following previous URC experience, the team has implemented significant modifications to each subsystem. The complete mechanical structure assembly is scheduled for January 10-12, 2025, with testing continuing until May 2025. Integration of autonomous systems is planned from December 2024 to January 2025, with testing continuing until late January. The control system undergone testing from December 2024 to February 2025.

c. Operational Testing: The operational testing phase runs from January to May 2025 and includes several key activities. From January 25 to May 10, 2025, the team will conduct autonomous testing. Extreme retrieval tests and ES tests are scheduled from mid-January to May 10, 2025. The science module will be set up on the rover in early February 2025, with controlled field tests scheduled for February 10-21, 2025. On-board science testing will run from February 25 to May 10, 2025. These tests ensure the rover can perform essential tasks including soil collection, sample analysis, switch and knob manipulation, cache box collection, and latch operations.



Figure – 2: End Effector

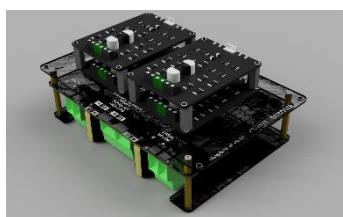


Figure – 3: Cytron Motor Driver

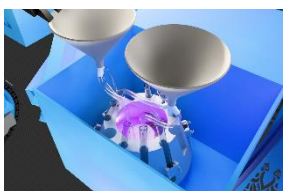


Figure - 5: Science Module



Figure – 6: Excavator Claw



Figure - 4: Hypersonic



Figure – 7: Panoramic View

ADD Gantt chart and budgets

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Science Mission:

The rover can satisfy the objective of SM in URC 2025. That are, to collect soil samples from sites by rover and perform onboard experiments of Nitrogen and Carbon by Molisch's reagent test, Ninhydrin test, and Ultraviolet-C (UV-C) analysis for detecting extinct, extant and fossilized life forms. At definite sites the operator uses a camera mounted on a servo motor to capture images every nine degrees, which are stitched together into 180-degree panoramas, given example in **Figure - 7**. Panoramic shots are stitched with indication of scale and cardinal direction is aligned with it. To determine cardinal directions, Pixhawk is implemented and gives raw data of pitch, yaw, and altitude. Using that data, a servo-based 78-degree FOV, an auto exposure, and an autofocus Logitech webcam generates a 180-degree panoramic view. By analyzing a geologist validates the results on site by using onboard camera using distance markers as well.

The selection for subsurface soil analysis is based on geological features observed such as humid soil samples or cracks on the surface, which indicates an easy drilling method. Soil condition is investigated using an NPK sensor which will pre-analyze the condition. By using an excavator claw depicted in **Figure- 6**, the rover can extract samples from 15 cm deep and channel the soil in two separate funnels leading to separate test tubes, which prevents cross contamination. A 3D printed cache box with a one-sided valve which avoids contamination, and a five-gm core sample is calculated by a load cell attached to the valve. The cache is detachable and will be presented to the judge for evaluation. Result is ensured by a combination of other sensors for atmospheric analysis such as carbon dioxide, air pressure, carbon monoxide, temperature, humidity and light intensity.

Presence of carbohydrates indicates extinct, extant or fossilized life form in soil samples. Onboard rover test utilizes Molisch's reagent to determine the presence of primary and tertiary carbohydrate sources. Soil samples are inserted in a test tube with pre-applied distilled water and Molisch's reagent (1:1) volume. The soil solution is centrifuged and α -Naphthol reacts with dehydrated carbohydrates representing the presence of simple carbohydrates. Peristaltic pump pumps the reaction to concentrated H_2SO_4 , which helps to form a purple ring indicating the presence of tertiary carbohydrate [1].

Protein helps in catalyzing nitrogen fixation that initiates abiotic life and other biochemical processes [2]. The rover uses a Ninhydrin test onboard where 10% ninhydrin solution is prepared and pre-loaded to the test tube with distilled water. After adding soil samples and centrifugation, the solution is mixed properly. Addition of heat by a nichrome wire coil inside the test tube for approximately three minutes results in formation of purple color in the test tube [3]. The results for both tests are observed by a color sensor that generates a graph in the GUI by which the result can be interpreted.

For co-relating the result of experiment from color-based analysis, further identification of carbohydrate and nitrogen is done by UV-C ray mathematical analysis. The team is trying to develop an auto generated response in GUI for proteins by molar absorption coefficient of UV-C due to the presence of tryptophan and tyrosine at 280 nm. UV absorbance of a protein depends on concentration and arrangement of these amino acids. For quantifying proteins UV absorbance is measured using a UV-C sensor, typically between 260 and 280 nm, making it ideal for pure proteins with tryptophan and tyrosine [4]. The limit of detection and quantification method is used to calculate the concentration of carbohydrate, depending on the value of the UV-C sensor by which the lowest analyte concentration is measured [5].

References:

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